

Maintenance Review Report: SMT-03 Reflow Oven Zone 5

Report No.: MNT-RPT-20260521-03

Owner: Maintenance Dept. Reviewer: Zhou Qiang

Event link: QE-20260521-001, Work order WO-260521-017, Sensor TEMP-05

Time window: 2026-05-21 09:12 to 09:35. Recipe: EC-CTRL-BGA-V2. Line: SMT-03.

Measured Temperature Profile

09:10 Zone5 actual 235.7 C vs target 239.0 C

09:15 Zone5 actual 232.6 C vs target 239.0 C

09:20 Zone5 actual 232.2 C vs target 239.0 C

09:35 Zone5 recovered to 237.5 C

No PLC hard alarm was triggered because the stop threshold was not reached.

Engineering Assessment

Finding 1: Zone 5 actual temperature stayed 6.8 C below recipe center for 11 minutes.

Finding 2: TEMP-05 calibration is valid until 2026-06-30, but connector resistance was above the internal warning limit.

Impact: The drift may increase solder void risk when solder paste warm-up time is short or cold-chain evidence is missing.

Immediate action: Run golden-board profile verification before restart and hold products processed from 09:00 to 10:00.

Corrective action: Replace TEMP-05 connector, tighten terminal block, and add weekly trend review for zones 4 to 6.

Release condition: Maintenance sign-off, quality X-Ray recheck, and production first-piece approval must be completed.